3-31-26 (WEEK 6)

Shun/翔海 (@shun4midx)

POISSON DISTRIBUTION QUANTITIES (INTUITION)

As we increase n (decrease size of subintervals), we expect each subinterval to have 0 or 1 events, with $\frac{\lambda}{n}$ on avg

Indep of events $\Rightarrow$ {whether or not a subinterval has an event}: indep

X = # events in unit interval $\approx$ # subintervals that have an event $\sim Bin(n, \frac{\lambda}{n})$

$$\therefore \mathbb{P}(X=k) = \lim_{n\to\infty} \mathbb{P}(Bin(n,\tfrac{\lambda}{n})=k) = \lim_{n\to\infty} \binom{n}{k}(\tfrac{\lambda}{n})^k(1-\tfrac{\lambda}{n})^{n-k} = \lim_{n\to\infty} \frac{\overbrace{n(n-1)\cdots(n-k+1)}^{\sim n^k}}{k!}(\tfrac{\lambda}{n})^k(1-\tfrac{\lambda}{n})^n(1-\tfrac{\lambda}{n})^{-k} = \frac{\lambda^k e^{-\lambda}}{k!}$$

DEFINITION

We define the probability mass function to be $\mathbb{P}(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$, $k=0,1,2,\ldots$

Claim: This is a valid distribution

Proof

$\mathbb{P}(X=k) \geq 0 \ \forall k$ ✓

$\sum_{k=0}^{\infty} \mathbb{P}(X=k) = \sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{-\lambda}\cdot e^{\lambda} = 1$ ✓ □

POISSON DISTRIBUTION QUANTITIES

Probability mass function: $\mathbb{P}(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k\mathbb{P}(X=k) = e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^k}{(k-1)!} = e^{-\lambda}\lambda\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda e^{-\lambda}e^{\lambda} = \lambda$$

$$Var(X) = \mathbb{E}[(X-\mathbb{E}(X))^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X^2-X+X] - \mathbb{E}[X]^2 = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2 = \mathbb{E}[X(X-1)] + \lambda - \lambda^2$$

↳ $\mathbb{E}[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\mathbb{P}(X=k) = \sum_{k=2}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\lambda^2\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{-\lambda}\lambda^2 e^{\lambda} = \lambda^2$

$\therefore Var(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$

REMARK

Recall, as $n\to\infty$, $Poi(\lambda) \approx Bin(n, \frac{\lambda}{n})$

↳ Expectation: $n(\frac{\lambda}{n}) = \lambda$

↳ Variance: $n(\frac{\lambda}{n})(1-\frac{\lambda}{n}) \longrightarrow \lambda$

} Based on Bin

REMARK (POISSON APPROXIMATION)

If n is large, and $np \to \lambda \in \mathbb{R}$: const, then we can approximate $Bin(n,p)$ by $Poi(np)$

THE POISSON PARADIGM

Suppose we have a probability space $(S, \mathbb{P})$, and a sequence of events $E_1, \ldots, E_n$, with $\mathbb{P}(E_i) = p_i \ \forall 1 \leq i \leq n$.

If the p_i are all "small", and the events are weakly independent $\leftarrow \mathbb{P}(E_i|E_j) \approx \mathbb{P}(E_i) \ \forall i \neq j$ or independent, then setting $\lambda = p_1 + p_2 + \ldots + p_n$, if X is the number of events that occur, then $X \approx Poi(\lambda)$

EXAMPLE

n people leaving a party, each takes a random hat. What is the probability that exactly k people get their own hat?

Solution

Define the events $E_i := \{i^{th}$ person gets their own hat$\}$

Then, $\mathbb{P}(E_i) = \frac{1}{n} \ \forall i \xrightarrow{n\to\infty} 0$

Notice, E_i aren't independent, but we claim they're weakly independent, since $\mathbb{P}(E_i|E_j) = \frac{1}{n-1} \approx \frac{1}{n} = \mathbb{P}(E_i)$

$\therefore$ By the Poisson paradigm, # people getting their own hat $\approx Poi(n\cdot\frac{1}{n}) = Poi(1)$, so $\mathbb{P}(X=k) \approx \frac{e^{-1}(1)^k}{k!} = \frac{1}{k!e}$

EXAMPLE

Shun/翔海(@shun4midx)

Toss a fair coin n times.

Let L_n = length of longest sequence of consecutive heads

What will L_n typically be?

<u>Solution</u>

$S = \{H, T\}^n$, $\mathbb{P}$ = uniform

$\{L_n \geq k\} = \{\exists \text{ seq of } \geq k \text{ heads in a row}\} =: E$

Here, $E := \bigcup_{i=1}^{n-k+1} E_i$, $E_i := \{\exists \text{ seq of } k \text{ heads in a row } \underline{\text{starting from toss } i}\}$

specify seq of len k

Notice, $\mathbb{P}(E_i) = \frac{1}{2^k}$: small if k is large if k is not too small.

We of course don't have independence, since streaks can overlap

Notice, $\mathbb{P}(E_{i+1} | E_i) = \frac{1}{2} >>> \mathbb{P}(E_i) \Rightarrow$ no weak indep (cannot use Poisson paradigm)

only need toss #(i+k) to be a head

We can try defining the events differently,

$E = \bigcup_{i=1}^{n-k+1} E_i'$, $E_i' = \{\exists \text{ seq of } k \text{ heads in a row, } \underline{\text{then ends with a tail or is the end of the seq}} \text{ starting from toss } \#i\}$

Here,

$$\mathbb{P}(E_i') = \begin{cases} 2^{-k-1}, & i \in [1, n-k] \quad \leftarrow \text{specify len } k+1 \text{ event} \\ 2^{-k}, & i = n-k+1 \quad \leftarrow \text{specify len } k \text{ event} \end{cases}$$

which is small.

$$\mathbb{P}(E_i' | E_j') = \begin{cases} \mathbb{P}(E_i'), & |i-j| > k+1 \\ 0, & |i-j| \leq k+1 \end{cases} \approx \mathbb{P}(E_i') \quad (\because \mathbb{P}(E_i') : \text{small})$$

$\therefore$ We have weak independence and Poisson paradigm applies

If $\underline{\lambda = \sum_{i=1}^{n-k+1} \mathbb{P}(E_i') = (n-k+2)2^{-(k+1)}}$, then $X_k := \#i \text{ s.t. } E_i' \approx Po(\lambda)$ $(n-k+2)2^{-(k+1)})$

Here, $\{L_n \geq k\} = \{X_k \geq 1\} \Leftrightarrow \{L_n < k\} = \{X_k = 0\}$

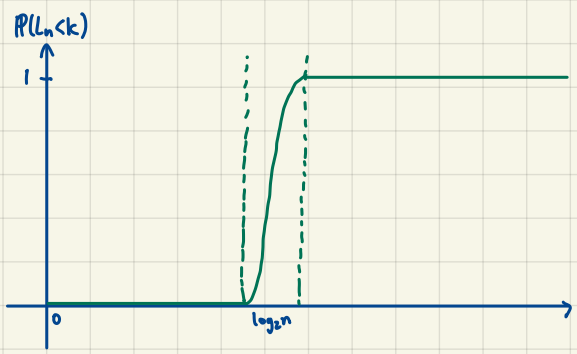
$\therefore \mathbb{P}(L_n < k) = \mathbb{P}(X = 0) \approx e^{-\frac{n-k+2}{2^{k+1}}}$

The "typical value" is when numerator $\approx$ denominator (if num > den, $\mathbb{P} \approx 0$, else, $\mathbb{P} \approx 1$)

If $k \leq (1+\varepsilon)\log_2 n$ for $\varepsilon > 0$, then $\frac{n-k+2}{2^{k+1}} \to \infty \Rightarrow \mathbb{P}(L_n < k) \to 0$

If $k \geq (1+\varepsilon)\log_2 n$, then $\frac{n-k+2}{2^{k+1}} \to 0 \Rightarrow \mathbb{P}(L_n < k) \to 1$

If $k = \log_2 n + C$ for some const c, $\frac{n-k+2}{2^{k+1}} \to 2^{-c} \Rightarrow \mathbb{P}(L_n < k) = e^{-2^{-c}} = \text{const}$



Shun/翔海(@shun4midx)

GEOMETRIC DISTRIBUTION

Parameter: Success probability $0<p\leq 1$, $X \sim Geom(p)$

Situation

Independent trials w/ success probability p

Count # trials until first success (inclusive)

Values: $k=1,2,3,4,\ldots$

GEOMETRIC DISTRIBUTION QUANTITIES

$$\mathbb{P}(X=k) = \underbrace{(1-p)^{k-1}}_{\text{first } k-1 \text{ fails}} \underbrace{p}_{\text{last success}} \geq 0$$

Notice, $\sum_{k=1}^{\infty} \mathbb{P}(X=k) = \sum_{k=1}^{\infty} p(1-p)^{k-1} = p \cdot \frac{1}{1-(1-p)} = 1$, so this is a valid probability ✓